

NudgeAI: AI-powered Context-aware Email Follow-up Agent

NudgeAI is a Workflow automation system using n8n, that can do the following:

- Track sent emails in Outlook
- Decides using AI if follow-up is needed and when to send follow-up
- Sends AI-generated natural sounding follow-up when no reply is received yet on follow-up send schedule

1. Definition of Terms

Parent Message - The original email sent by the sender that is awaiting a reply from the recipient.

Sender - The email address or user account that sends the parent message. Serves as the owner of the system and the connections needed in n8n.

Recipient - The email address or user account that receives the parent message.

Reply Message - The email sent by the recipient as a response to the parent message.

Follow-up Message - The email sent by the sender as a follow-up to the parent message when no reply message is detected within the defined time frame.

Test Environment - Environment for testing the flow. This will let NudgeAI send follow-up immediately after detecting a new message saved in its database. Activated by setting 'isActualEnvironment' variable in Config workflow to false.

Actual Environment - Environment for actual usage. This will let NudgeAI send follow-up based on the schedule devised by Agent. Activated by setting 'isActualEnvironment' variable in Config workflow to true.

2. Concept

Figure 1 shows the typical email follow-up flow. This serves as the template for any existing email follow-up tools out there.

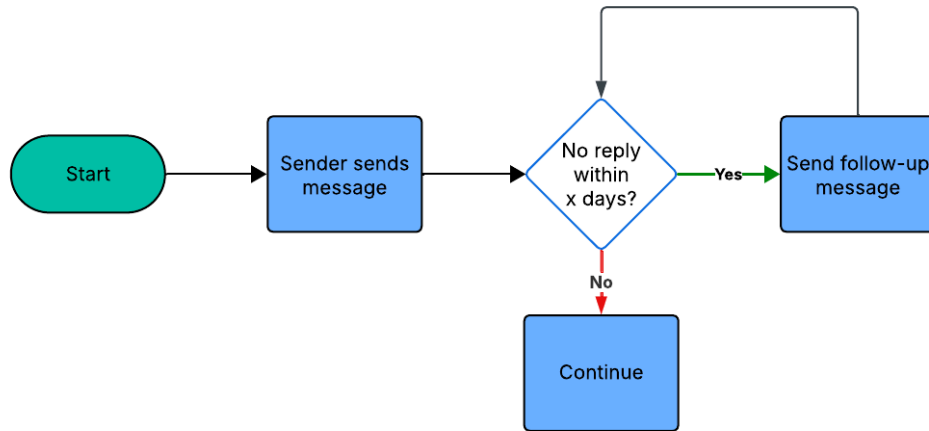


Figure 1. Typical Email Follow-up flow

This would mean that different tools differ only in the way decisions and values are made. Table 1 shows the difference between existing Email tools, and this proposed AI-powered system.

Decision Point	Existing Email follow-up tools	NudgeAI
Is Follow-up necessary?	Set manually for each email	Set automatically using contextual cues
Days before follow-up	Set manually for each email	Set automatically using contextual cues
Creation of Follow-up message	Uses static predefined templates or generic messages	Generates dynamic , context-aware messages tailored to conversation tone and content

Table 1. Difference between existing tools and NudgeAI

Table 1 shows that NudgeAI makes use of the context of the email to make decisions in following up. This would mean that NudgeAI is ideal for entities that are expected to handle multiple emails, such as Customer Support Helpdesks, in which manual decision-making is just not efficient. Further details about this usage and scope of the context used is explained in the next section.

3. Key Features

Baseline features:

These are features that Email follow-up tools have.

- Monitoring of sent emails and received replies
- Automatic sending of a follow-up message

Differentiators:

These are the features that differentiate NudgeAI from existing tools.

Deciding the necessity of follow-up message

NudgeAI decides if a follow-up message is required or not based on the content of the original message.

Situations where follow-up is not necessary:

- Parent message is gibberish
- Parent message is purely informational
- Parent message explicitly says to not reply

Context-based Scheduling

NudgeAI uses 4 factors with types defined as the following:

1. Urgency:
 - high: blocking issue, customer problem, revenue-impacting
 - medium: expected response but not critical
 - low: optional or informational
2. Relationship:
 - internal: same company
 - external: customer/client
 - VIP: high-value client or priority account
 - cold: no prior relationship or outreach
3. Nature of the email:
 - support: resolving an issue/problem
 - sales: trying to close or move forward
 - info: providing information only
 - decision: waiting for approval or confirmation

4. Time Sensitivity:

- immediate: action expected within hours
- short: action expected within 1–3 days
- long: action expected within a week or more
- none: no clear deadline

Context-based Writing

The same 4 factors above are used by NudgeAI to write follow-up messages using context. Furthermore, it is set to dynamically write with natural sounding words and also follows the style and tone of the parent message. All of these key features are made possible due to a detailed rule-based prompt that is shown in the **Implementation** section.

4. Test Environment and Results

Note: Testing was done in the Test Environment, which is briefly explained in the Definition of Terms section, and is detailed in the Implementation section.

Roles:

Sender: Virgel - virgelraitorres@outlook.com

Recipient: Rai - torresrai618@outlook.com

Scenario 1: Sender sent something, but no reply received

Subject:

Required Submission of Updated list of items

Body:

Good day Rai,

I'm reaching out to request the updated list of items to be recorded. Could you please share the latest version by tomorrow?

Kindly follow the format shown in <http://somewebsite.com>.

This will help us keep everything aligned and ensure smooth coordination moving forward.

Thanks,
Customer Service

Results:

Urgency: high Relationship: external Nature: decision Time Sensitivity: short Should Follow Up: true Recommended Delay In Days: 1 Follow-up Message: Good day Rai,/n/nJust a friendly reminder regarding the updated list of items to be recorded that we requested yesterday. The deadline for sharing the latest version is today./n/nKindly follow the format shown in __URL_0__/n/nSharing this promptly will help us keep everything aligned and ensure smooth coordination moving forward./n/nThanks,/n/nCustomer Service, Reason: The original email requests a 'Required Submission of Updated list of items' with a clear deadline of 'by tomorrow' (May 23rd from the May 22nd Sent Time), indicating high urgency and short time sensitivity. The request is for a specific deliverable crucial for 'smooth coordination'. A follow-up is necessary to ensure the list is provided. Setting a recommended delay of 1 day from the current time means the follow-up will be sent on May 23rd, serving as a timely reminder on the deadline day. The message reflects this by referring to the request from 'yesterday' and the deadline as 'today'.	<pre>output urgency : high relationship : external nature : decision timeSensitivity : short shouldFollowUp : true recommendedDelayInDays : 1 followupMessage : Good day Rai,/n/nJust a friendly reminder regarding the updated list of items to be recorded that we requested yesterday. The deadline for sharing the latest version is today./n/nKindly follow the format shown in __URL_0__/n/nSharing this promptly will help us keep everything aligned and ensure smooth coordination moving forward./n/nThanks,/n/nCustomer Service reason : The original email requests a 'Required Submission of Updated list of items' with a clear deadline of 'by tomorrow' (May 23rd from the May 22nd Sent Time), indicating high urgency and short time sensitivity. The request is for a specific deliverable crucial for 'smooth coordination'. A follow-up is necessary to ensure the list is provided. Setting a recommended delay of 1 day from the current time means the follow-up will be sent on May 23rd, serving as a timely reminder on the deadline day. The message reflects this by referring to the request from 'yesterday' and the deadline as 'today'.</pre>
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Table 2. NudgeAI's decision output for Scenario 1

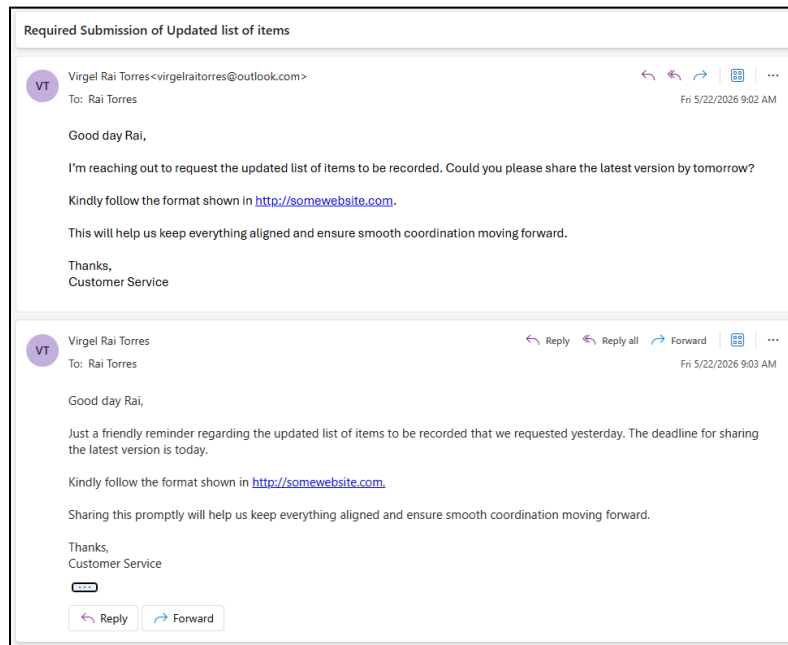


Figure 2. (Sender view) Parent message and Follow-up message shown

Scenario 2: Sender sent something, recipient replied

Parent Message:

Subject: Application Form to be filled out

Body:

Hi Rai,

To process your application, I'd like to request you to fill out the application form attached in this email. Once received, I'll confirm and proceed with the review.

Looking forward to your submission.

Thank you,
Virgel

Reply Message:

Body:

Good morning Virgel,

Attached here is my submission for the application form as requested. Please let me know if there are things that need to be done on my end.

Thank you,
Rai

Results:

No follow-up message was sent by NudgeAI.

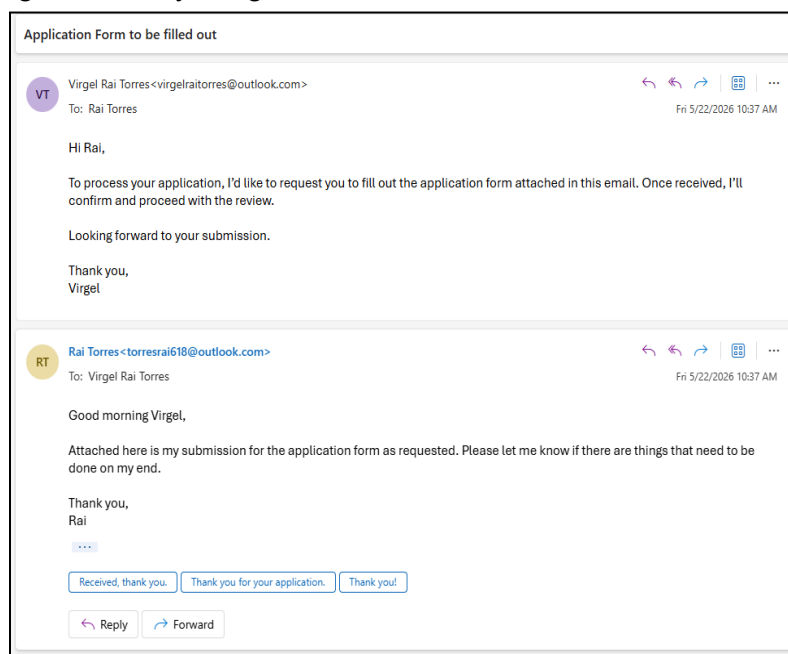


Figure 3. (Sender view) Parent message and Reply message shown

Scenario 3: Sender sent something informational, no reply needed

Subject:

Ticket # 172384627 resolved

Body:

Good day,

Just want to inform you that everything regarding your concern in this Ticket #: 172384627 is resolved.

Thank you.

Results:

Urgency: low Relationship: external Nature: support Time Sensitivity: none Should Follow Up: false Recommended Delay In Days: 0 Follow-up Message:No follow-up is recommended as this email confirms the resolution of a support ticket. Reason: The original email confirms that Ticket # 172384627 has been resolved. As per the rules, follow-ups are not to be sent solely to confirm if a resolution is satisfactory; the recipient will normally reply if they encounter an issue.	output urgency : low relationship : external nature : support timeSensitivity : none shouldFollowUp : false recommendedDelayInDays : 0 followupMessage : No follow-up is recommended as this email confirms the resolution of a support ticket. reason : The original email confirms that Ticket # 172384627 has been resolved. As per the rules, follow-ups are not to be sent solely to confirm if a resolution is satisfactory; the recipient will normally reply if they encounter an issue.
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Table 3. NudgeAI’s decision output for Scenario 3

5. Implementation

Database used:

NudgeAI uses an **Excel spreadsheet** as a way to record data for tracking and processing. Any form of database, as long as supported by n8n can be used in actuality, however Excel was used due to its modularity and simplicity in terms of preparing and setting up. The Excel workbook simply serves as a recording tool and therefore should not be edited manually. The structure used by NudgeAI is shown below.

Template:

Workbook

- Sheet name
 - Table name
 - Column1, Column2, ...
- Sheet name
 - Table name
 - Column1, Column2, ...

Actual:

Workbook

- Emails
 - emails
 - messageId, conversationId, subject, recipient, sender, sentTime, status, shouldFollowUp, nextFollowUpAt, followUpMessage, repliedTime, urgency, relationship, nature, timeSensitivity, recommendedDelayInDays, reason
- Metadata
 - metadata
 - id, allLastCheckedAt

The metadata table only has one row, with id set to 1, used to record when the last time sent emails were checked by NudgeAI.

State transitions:

NudgeAI records the history and status of the processing of emails through state transitions. All status, except “New” are reflected in the Excel DB. Figure 4 shows the State Transition Diagram used.

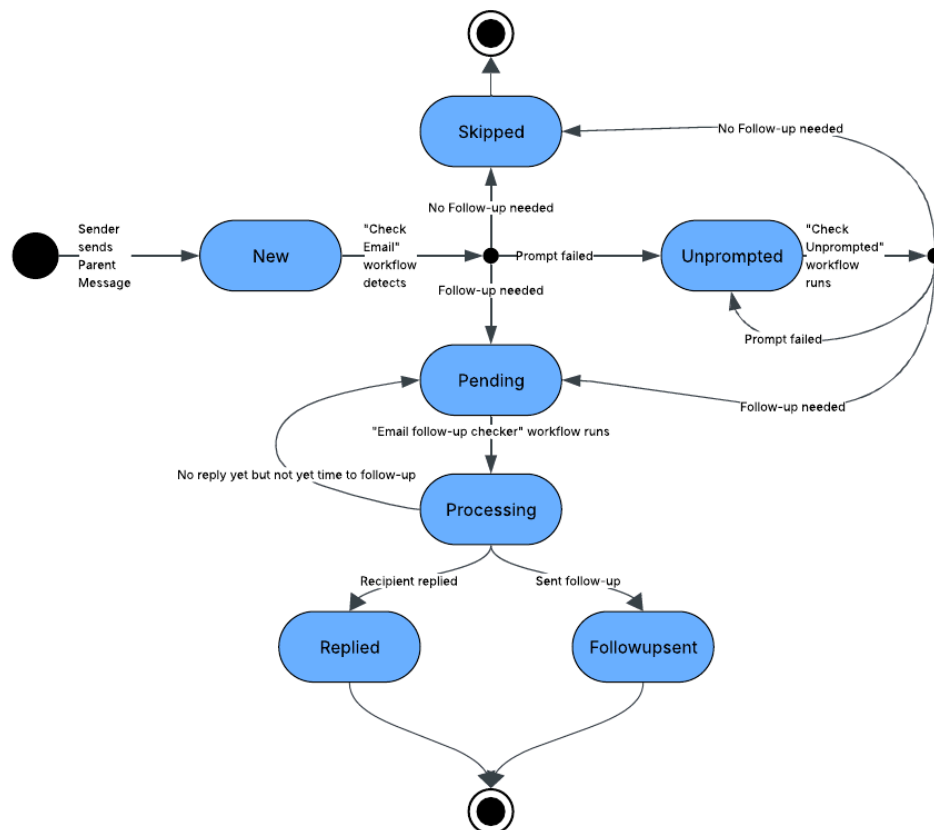


Figure 4. State Transition Diagram

Workflows structure:

To implement the desired functionality of NudgeAI, the entire process can be divided into 2 main parts, namely:

- Detection and tracking of sent emails
- Handling of follow-up and reply cases

In terms of state transition, the first part ends in either skipped or pending state, while the second part always starts from pending state.

The first part consists of 3 major workflows, with their interaction shown in Figure 5.

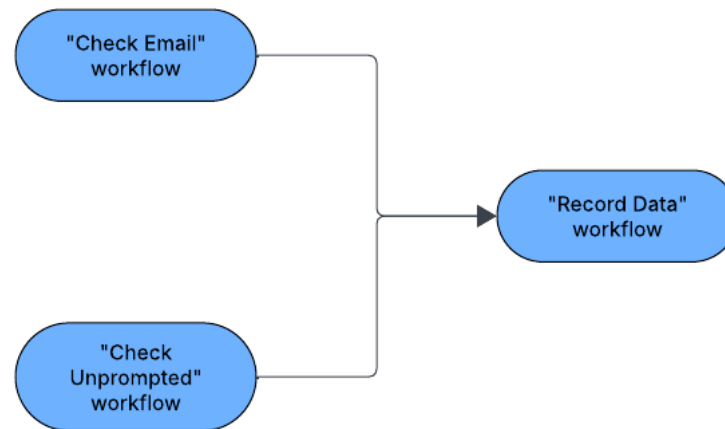


Figure 5. Workflows for Detection and tracking of Sent emails

The second part only uses one workflow “Email follow-up checker”.

Workflows:

Workflows are categorized into 3, namely, Main, Helper, and Config. Main workflows act as gateways to the process and are set to be triggered to run on regular intervals. Helper workflows act as sub-workflows that multiple Main and helper workflows can call, and also to test critical functions in isolation. Finally, there is only one config workflow that is used to set the link to the Excel DB, and the environment, whether Test or Actual.

Main Workflows

1. Check Email

Check Email is done with 4 major steps, which are the following:

- Get messages sent by sender on or after it was last checked. (allLastChecked)
- Identify Parent messages of each of the emails obtained.
- Pass Parent messages to “Record Data” workflow
- Update allLastCheckedAt to the time the workflow if the DB has been updated.

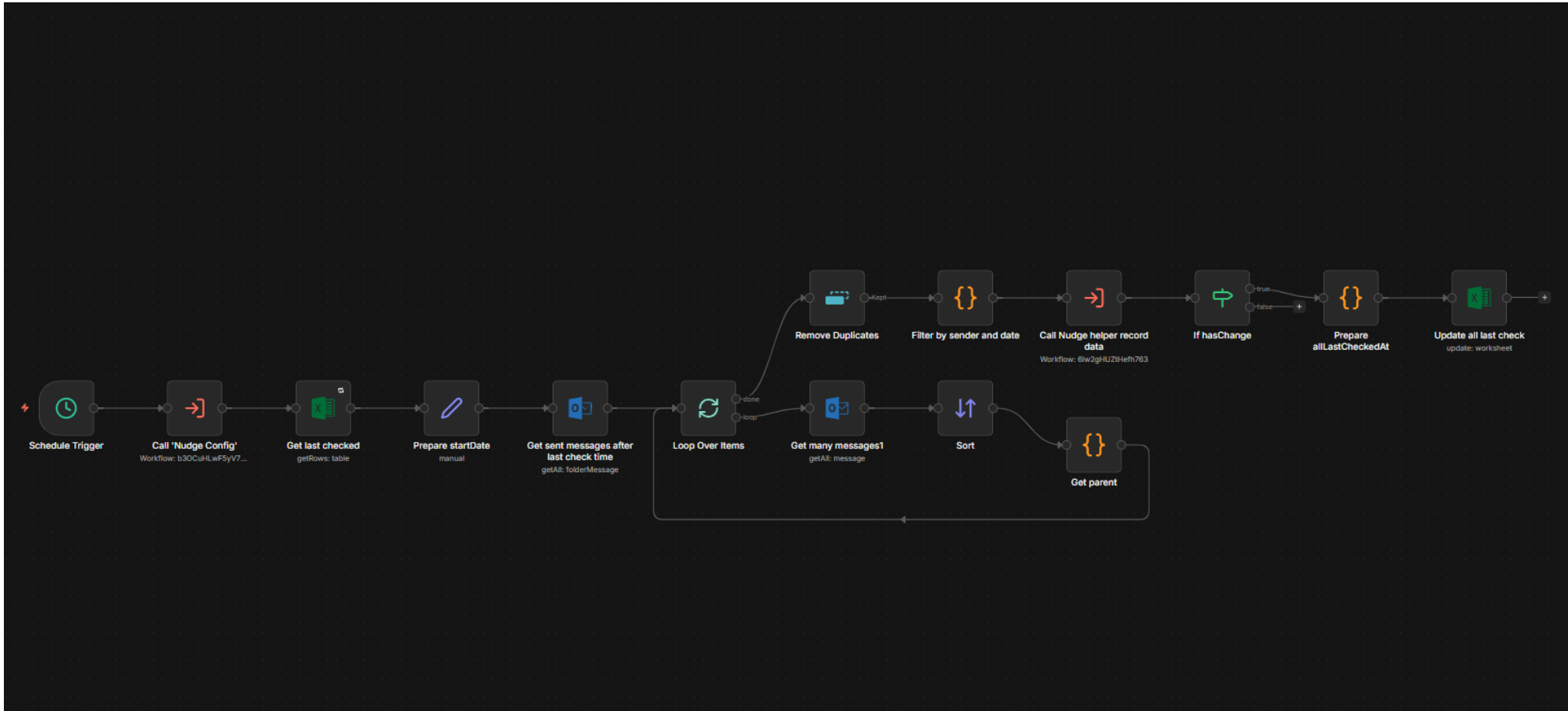


Figure 6. “Check Email” Main workflow

2. Check Unprompted

Check Unprompted workflow gets the rows in the emails table with “unprompted” status, then using the messageId, get the original message for its metadata, then pass these messages to “Record Data” workflow.

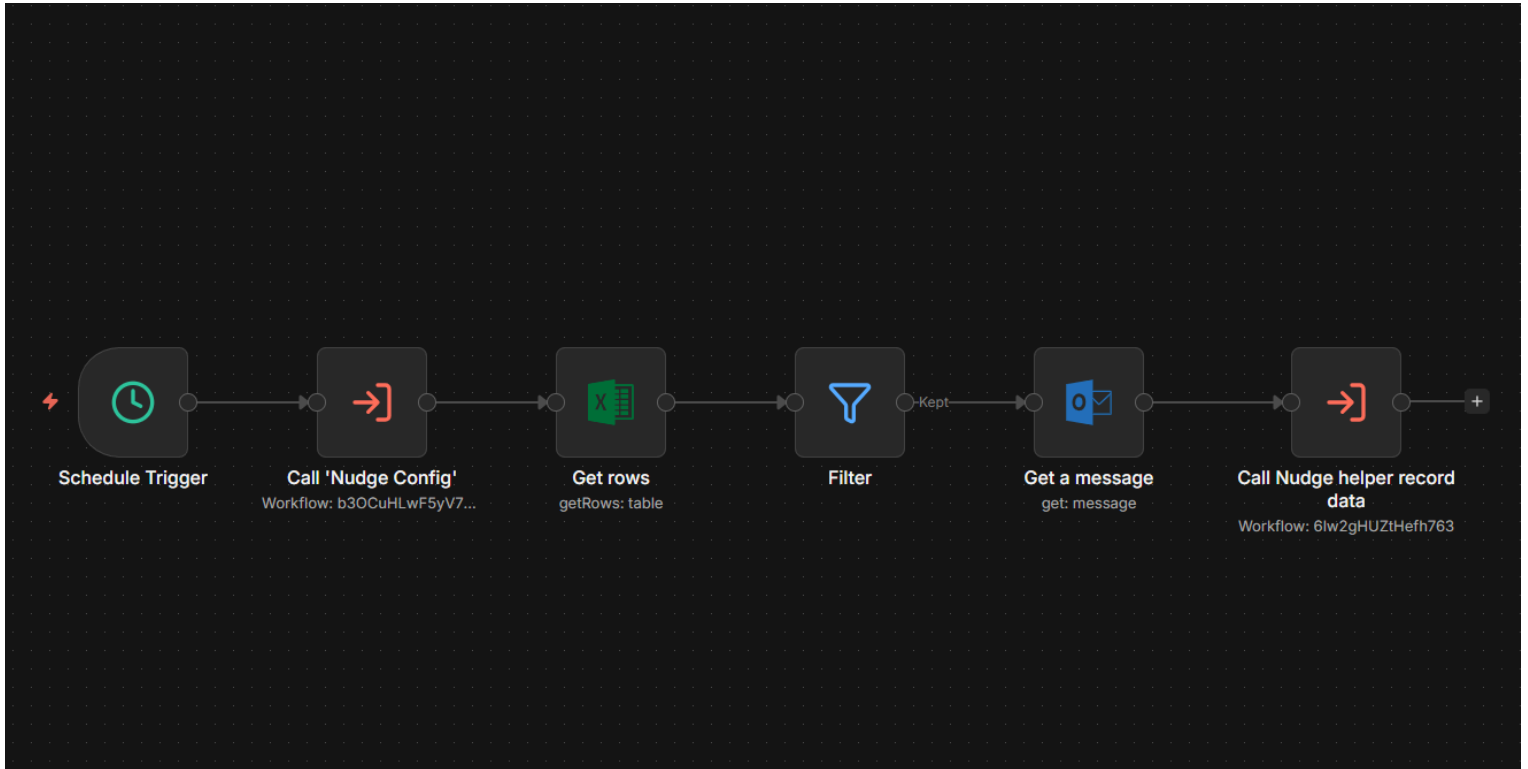


Figure 7. “Check Unprompted” Main workflow

3. Email followup checker

This workflow gets rows in the Emails table with pending status, then checks if it is time to follow up. If so, it uses both messageId, and conversationId to check if there are replies from the recipient. It sends the follow-up message if there is no reply. Finally, it updates the status in the table accordingly.

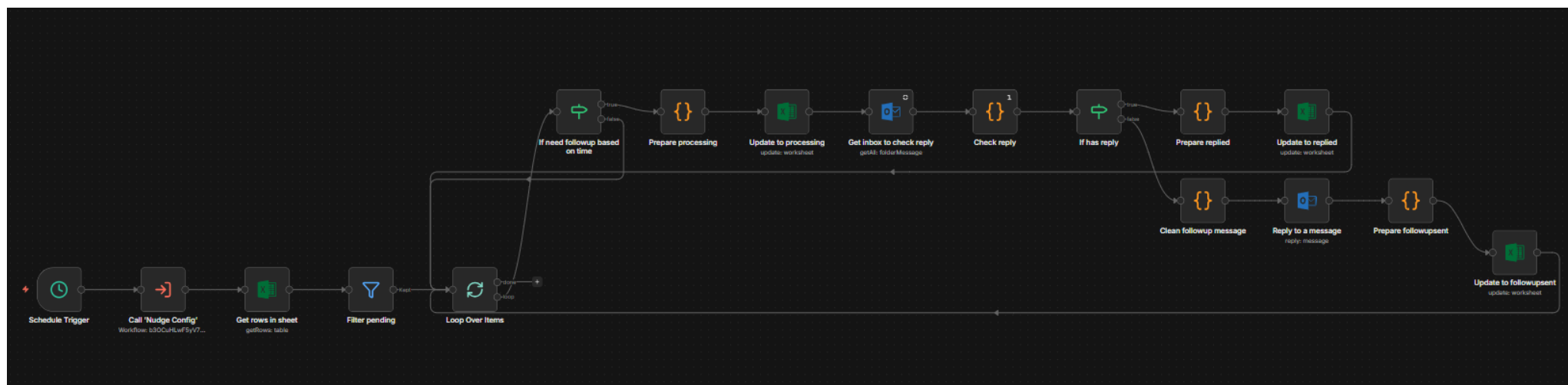


Figure 8. “Email followup checker” Main workflow

Helper Workflows

1. Followup prompt

This is the AI node for the decision making logic using the factors described in the Key Features section.

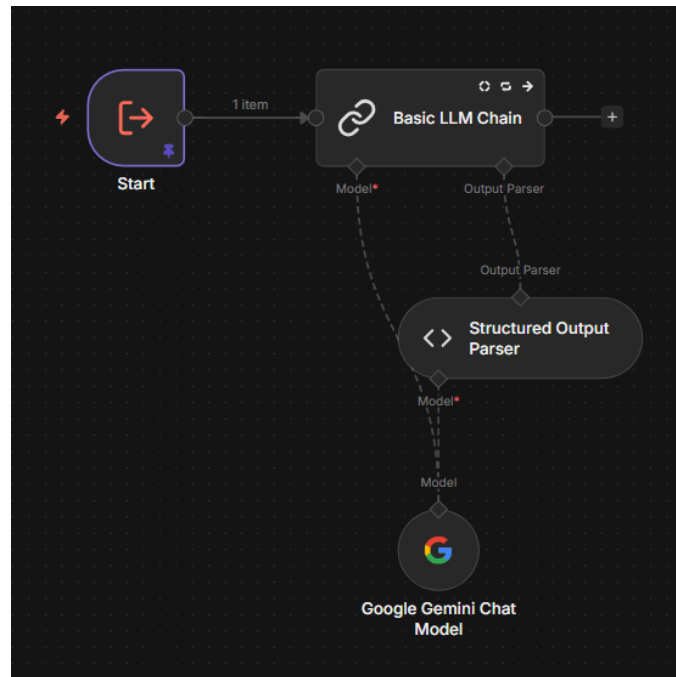


Figure 9. “Followup prompt” Helper workflow

Shown below is the prompt of NudgeAI, which gives a detailed prompt with rules to control how the AI makes the decisions.

You are an AI assistant that determines when a follow-up email should be sent and the followup message. The followup message is to be sent after recommendedDelayInDays from Current Time, but the content is based from Sent Time, so the content should feel like so. Use the redacted values wisely. Copy the same format as the original body but replace newlines with '\n'.

Analyze the email context and classify it based on the following factors:

1. Urgency:

- high: blocking issue, customer problem, revenue-impacting
- medium: expected response but not critical
- low: optional or informational

2. Relationship:

- internal: same company
- external: customer/client
- VIP: high-value client or priority account
- cold: no prior relationship or outreach

3. Nature of the email:

- support: resolving an issue/problem
- sales: trying to close or move forward
- info: providing information only
- decision: waiting for approval or confirmation

4. Time Sensitivity:

- immediate: action expected within hours
- short: action expected within 1–3 days
- long: action expected within a week or more
- none: no clear deadline

Then determine:

- whether a follow-up is needed (true/false)
- recommended delay before follow-up (in hours or days)

Rules:

- Do NOT recommend follow-up if the email is gibberish or is purely informational with no action required regardless if external or internal.
- Do NOT follow-up if the email explicitly states to only let them know if the recipient encountered an issue.
- Do NOT follow-up just to confirm if resolution is satisfactory. The recipient will normally just reply by themselves if they find an issue.
- Prioritize faster follow-ups for high urgency + external/VIP
- Avoid aggressive follow-ups for low urgency or internal messages
- Ensure that the followupMessage tone matches the recommendedDelayInDays.
 - shorter delay -> lighter nudge
 - longer delay -> stronger follow-up

CRITICAL RULE:

Do NOT modify placeholder tokens such as __URL_0__, __EMAIL_0__, etc.

- Do not rename them
- Do not partially rewrite them
- Do not add/remove characters
- Always copy them EXACTLY as provided

Return your answer in JSON format:

```
{
  "urgency": "high",
  "relationship": "external",
  "nature": "support",
  "timeSensitivity": "short",
  "shouldFollowUp": true,
  "recommendedDelayInDays": 1,
  "followupMessage": "...",
  "reason": "..."
}
```

Email:

Sent Time: {{ \$json.sentTime }}

Current Time: {{ new Date().toISOString().replace(/\.d{3}Z\$/, 'Z') }}

{{ \$json.subject }}

{{ \$json.body }}

2. Record data

This is responsible for adding or updating a row in the Emails table in Excel, the status results to one of the following: pending, skipped, and unprompted depending on the result of the Followup prompt.



Figure 10. "Record data" workflow

Config Workflow

1. Config

This is where you can set up the Excel file using the Excel ID, as well as the environment.

The Set Configs node has two variables:

- excelid
- isActualEnvironment

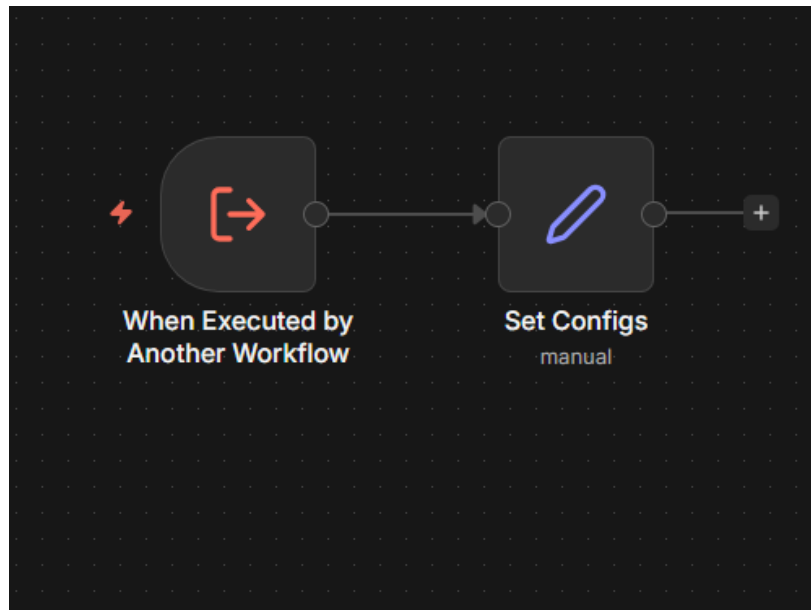


Figure 11. “Config” workflow